STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: FAB-03 | TAG: 01166873

Tennsmith HBU48-16 Hand Brake & Industrial Metal Shear

Facility Zone: Zone 5: Fabrication, Sheet Metal & Abrasive Finishing (FAB) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Glasses
2. Cut-Resistant Leather Gloves for Handling Sharp Sheet Metal Edges
3. Steel-Toe / Closed-Toe Shoes

2. PRIMARY OPERATIONAL HAZARDS

1. Tennsmith HBU48-16 Brake: 300 ft-lb counterweight swing and finger pinch hazard under box fingers
2. Industrial Shear: Guillotine blade amputation hazard behind hold-down guard

3. BEFORE OPERATION CHECKLIST

1. Never exceed rated mild steel capacity (16 gauge max on Tennsmith HBU48-16; never shear hardened steel, rod, or wire).
2. Verify all box brake fingers are tightly bolted before bending.

4. DURING OPERATION SAFETY RULES

1. Keep fingers completely clear of clamping nose bar and shear hold-down feet.
2. Keep bystanders clear of the 300 ft-lb counterweight arc when lifting bending leaf.

5. AFTER USE & SHOWROOM CLEANUP

1. Lower bending leaf, power down Shear, and place sheet metal offcuts in scrap bin.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Push red stop button on Shear immediately; release clamping lever on Hand Brake.